

Systematic Evaluation of Signal-to-Image Transformation Pipelines for Vision Transformer-based Sleep Stage Classification

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Abstract

Sleep stage classification is a critical prerequisite for diagnosing sleep disorders and assessing sleep quality. While Vision Transformers (ViTs) have demonstrated remarkable success in image recognition, existing ViT-based approaches to polysomnography (PSG) sleep staging rely mainly on spectrogram or raw-signal inputs, and the optimal transformation of one-dimensional physiological signals into image representations suitable for ViT processing has not been systematically investigated. This paper presents a comprehensive evaluation of four image transformation techniques - Gramian Angular Summation Field (GASF), Gramian Angular Difference Field (GADF), Markov Transition Field (MTF), and spectrogram - for ViT-based sleep stage classification using the CAP Sleep Database (80 recordings, 80,667 epochs) with subject-wise 5-fold cross-validation. Experimental results demonstrate that GASF achieves the highest classification accuracy at 78.5% among the four transformations. Furthermore, Discrete Wavelet Transform (DWT) down sampling at 128 Hz optimizes performance, yielding 80.3% accuracy. Fine-tuning an ImageNet pre-trained ViT-B/16 model on balanced GASF representations (class balance obtained through DCGAN-based synthetic augmentation) achieves 86.2% accuracy (macro-F1 0.853), representing an improvement of 11.4 percentage points over training from scratch and 10.0 percentage points over a ResNet-50 baseline. These results indicate that GASF encoding with DWT resampling to 128 Hz is an effective preprocessing pipeline for ViT-based sleep stage classification, and that transfer learning from natural image pre-training is highly effective for PSG-derived representations.

Keywords: Sleep stage classification; Vision Transformer, Gramian Angular Field; Signal to image transformation, Transfer learning.

Introduction

Sleep disorders represent a significant global health burden, affecting an estimated 50 - 70 million adults in the United States alone, with conditions such as insomnia, sleep apnea, and narcolepsy contributing to reduced quality of life, impaired cognitive function, and increased cardiovascular and metabolic risks (Hirshkowitz et al., 2015; Itani et al., 2017). Accurate sleep stage classification forms the foundation of clinical sleep assessment, enabling physicians to diagnose disorders and evaluate treatment efficacy. The gold standard relies on manual scoring of polysomnography (PSG) recordings by trained technologists following standardized criteria; however, this process is labour-intensive, requiring approximately 2 - 4 hours per full-night recording, and suffers from inter-rater variability reaching only 82.6% even among experienced scorers (Rosenberg & Van Hout, 2013). These limitations have motivated extensive research into automated classification using deep learning, with early approaches employing Convolutional Neural

Networks (CNNs) for spatial feature extraction and Recurrent Neural Networks (RNNs) for temporal dependencies. The subsequent emergence of the Transformer architecture, with its self-attention mechanism capable of modelling long-range dependencies, has opened new avenues for sleep staging, particularly through Vision Transformers (ViTs) that process image representations of physiological signals (Vaswani et al., 2017; Dosovitskiy et al., 2021).

A fundamental requirement for applying ViTs to PSG data is the transformation of one-dimensional time-series signals into two-dimensional image representations. Several encoding strategies have been proposed, including Gramian Angular Summation Field (GASF), Gramian Angular Difference Field (GADF), Markov Transition Field (MTF), and spectrograms based on the Short-Time Fourier Transform (STFT) (Wang & Oates, 2015). While these methods capture distinct aspects of the underlying signal - temporal correlations, state transitions, or time-frequency characteristics - existing ViT-based sleep studies predominantly employ spectrograms or raw one-dimensional signals, leaving the comparative effectiveness of GASF, GADF, and MTF for ViT processing entirely unexplored. Furthermore, the optimal sampling frequency and down sampling method for PSG signals prior to image transformation have not been systematically investigated, and the effectiveness of transfer learning with pre-trained ViTs for PSG-derived images remains insufficiently characterized, as most existing models are trained from scratch. This study addresses these gaps by systematically evaluating image transformation pipelines for ViT-based sleep stage classification on the CAP Sleep Database.

Several recent studies have applied Vision Transformers to spectrogram-based sleep staging, though none have systematically compared GASF, GADF, and MTF encodings for this purpose. Peng et al. (2023) proposed a patch-based spectrogram Transformer pre-trained on image and audio data (accuracy gains of 1.8-2.5 percentage points on Sleep-EDF); Ardeshtir and Araghi (2023) reached 86.97% on Sleep-EDF-20 with a pre-trained CNN on spectrograms; Guo et al. (2024) reported 81.83% on a clinical cohort of sleep-disorder patients (SleepUHS); and Lee et al. (2025) obtained a macro-F1 of 81.94% with a ViT emulating human visual scoring. These studies rely on spectrogram or raw-signal input and mostly on healthy or lightly disordered cohorts, leaving open which time-series-to-image encoding best suits ViT self-attention, and how that choice interacts with resampling, pre-training, and architecture on a heterogeneous cohort such as CAP.

Materials and Methods

Dataset and Data Collection

The experiments utilized the CAP Sleep Database, publicly available from PhysioNet (Terzano et al., 2001; Goldberger et al., 2000). This database contains 108 full-night polysomnographic (PSG) recordings acquired at the Sleep Disorders Center of Ospedale Maggiore di Parma, University of Parma, Italy. Each recording includes 16 bipolar channels sampled at 512 Hz: 12 electroencephalography (EEG) derivations (F7-T3, T3-T5, Fp1-F3, F3-C3, C3-P3, P3-O1, Fp2-F4, F4-C4, C4-P4, P4-O2, F8-T4, T4-T6), bilateral electrooculography (EOG), submental electromyography (EMG), electrocardiography (ECG), and an averaged EEG reference signal. Sleep stages were scored by expert technicians according to Rechtschaffen and Kales (R&K) criteria, yielding labels for Wake, S1, S2, S3, S4, and REM. Following a subject selection protocol, 80 recordings were retained, encompassing 6 healthy subjects and 74 patients with various sleep disorders. The full dataset comprises 80,667 epochs after exclusion of movement artifacts and unscored segments. Table 1 presents the epoch distribution across sleep stages. Model evaluation used subject-wise 5-fold cross-validation over the full cohort: folds were assigned by subject, so no subject's epochs appeared in both training and test data, and each iteration trained on four folds and tested on the fifth with a fixed training schedule (see Experimental Design and Training Procedure) and no validation subset or early stopping. Tables 2-5 give the mean $\pm$ standard deviation of test-set performance across the 5 folds.

Class balance for the balanced-data conditions (Tables 4 and 5) was achieved with DCGAN-based synthetic image augmentation: a class-conditional Deep Convolutional GAN was trained to generate additional synthetic epoch images for minority-class sleep stages, bringing each class up to the majority class count. Both GAN training and generation used only the four training folds of each cross-validation iteration, so no synthetic epoch was influenced by, or could appear in, the test fold. GAN architecture and fidelity metrics are given in Appendix A.

Signal Preprocessing

Each of the 16 PSG channels was preprocessed independently. A fourth-order Butterworth high-pass filter at 0.3 Hz removed baseline wander and DC drift, while a low-pass filter at 45 Hz eliminated high-frequency components beyond the relevant EEG band. A 50 Hz notch filter suppressed power-line interference. Independent Component Analysis (ICA) was applied using the Infomax algorithm (Makeig et al., 1996) to identify and remove ocular, muscular, and cardiac artifact components. Signals were then down sampled to target frequencies (256 Hz, 128 Hz, or 64 Hz) using finite impulse response (FIR) low-pass filtering with appropriate anti-aliasing cutoffs.

Table 1: Epoch distribution across sleep stages in the CAP Sleep Database

Sleep Stage	Epoch Count	Percentage (%)
Wake	10,853	13.5
S1	5,227	6.5
S2	38,440	47.7
S3	8,467	10.5
S4	6,973	8.6
REM	10,707	13.3
Total	80,667	100.0

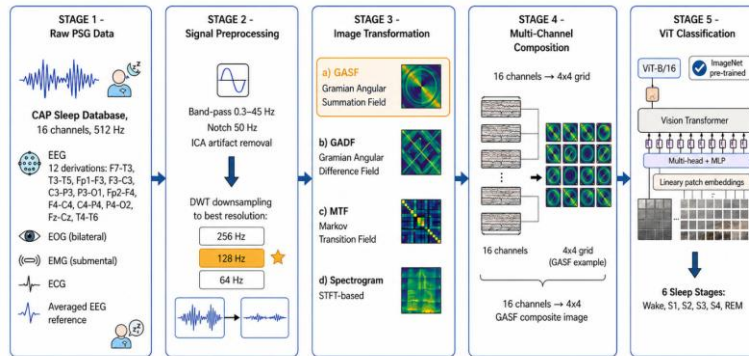


Figure 1: Overall pipeline of the proposed system. Raw 16-channel PSG signals are (Stage 1) acquired at 512 Hz, (Stage 2) filtered (0.3-45 Hz band-pass, 50 Hz notch, ICA artifact removal) and down sampled by FFT or DWT to 256, 128, or 64 Hz, and (Stage 3) converted by each of four image transformations (GASF, GADF, MTF, spectrogram). Stage 4 composes the 16 single-channel images into a 4x4 grid (224x224x3 composite image), and Stage 5 classifies each 30-second epoch into one of six sleep stages with an ImageNet pre-trained ViT-B/16.

Image Transformation Methods

Four image transformation techniques were evaluated for converting one-dimensional time series into two-dimensional image representations. All methods operate on a normalized time series $\tilde{\mathbf{x}} = [\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_N]$ where each element is scaled to $[-1, 1]$ via min-max normalization.

The GASF and GADF encodes temporal correlations through the cosine of summed and difference angular projections. Each normalized value $\tilde{x}_i$ is mapped to an angular coordinate $\theta_i = \arccos(\tilde{x}_i)$. The GASF and GADF matrix are computed as:

$$\text{GASF}(i, j) = \cos(\theta_i + \theta_j)$$

$$\text{GADF}(i, j) = \sin(\theta_i - \theta_j)$$

The MTF constructs a probabilistic representation by quantizing the time series into Q discrete bins and computing state transition probabilities. Let $q_i \in \{1, \dots, Q\}$ denote the quantized state of $\tilde{x}_i$, and let $\mathbf{P}$ denote the $Q \times Q$ first-order Markov transition matrix where $P_{mn} = P(q_j = n \mid q_i = m)$. The MTF is define as:

$$\text{MTF}(i, j) = P(q_j \mid q_i)$$

Following Wang and Oates (2015), $Q = 32$ bins were used. The spectrogram was computed via the Short-Time Fourier Transform (STFT) using a Hamming window of length 256 samples with 50% overlap.

Multi-Channel Image Composition and Model Architecture

Each of the 16 preprocessed PSG channels was independently transformed into a two-dimensional image. To construct a unified input compatible with standard ViT architectures, the 16 single-channel images were arranged in a 4×4 grid and concatenated along the channel dimension. The composite image was resized to 224×224×3 via bicubic interpolation, matching the expected dimensions of ImageNet-pretrained ViT models. Each position in the 4×4 grid consistently corresponded to a specific PSG derivation across all epochs.

The primary model was ViT-Base with 16×16 patch size (ViT-B/16), comprising 12 Transformer encoder layers, a hidden dimension of 768, 12 self-attention heads, and an MLP dimension of 3072. The input image was partitioned into 196 non-overlapping 16×16 patches, each linearly embedded into a 768-dimensional vector. A learnable class token and sinusoidal position embeddings were prepended. The classification head was replaced with a linear layer projecting to 6 sleep stage classes.

Experimental Design and Training Procedure

Four systematic experiments were conducted: (1) comparison of the four image transformation methods using pre-trained ViT-B/16 at 256 Hz; (2) evaluation of sampling frequency (256 Hz, 128 Hz, 64 Hz) using both Fast Fourier Transform (FFT) and Discrete Wavelet Transform (DWT) down sampling; (3) comparison of fine-tuned ImageNet weights versus training from random initialization under balanced and imbalanced conditions; and (4) comparison of seven architecture and scale variants: ViT-B/16, ViT-B/32, DeiT-B, Swin-B, BEiT-B, PVT-Small, and a CNN baseline (ResNet-50) - spanning patch size, model scale, pre-training objective, and a convolutional reference point.

All experiments were conducted on a workstation with an NVIDIA RTX A5000 GPU (24 GB VRAM). The software environment comprised PyTorch 2.0 and the HuggingFace Transformers library. Models were trained using the AdamW optimizer with an initial learning rate of 10^{-4} , weight decay of 0.01, and cosine annealing scheduling over 50 epochs with batch size 32. For from-scratch training, the learning rate was 3×10^{-4} and training extended to 100 epochs.

Evaluation Metrics

Model performance was assessed using overall accuracy (ACC) and macro-averaged F1-score. All metrics were computed on the held-out test set as mean $\pm$ standard deviation across the subject-wise 5-fold cross-validation described above. Headline accuracy and macro-F1 (Tables 4 and 5) are means of five per-fold values, whereas the per-class metrics in Appendix Table A1 and the confusion matrices in Appendix

Tables A2 and A3 are computed from pooled out-of-fold predictions over the full 80,667-epoch cohort. The two conventions can differ when folds are unequal in size, as under subject-wise folding over CAP's unevenly sized subgroups: macro-F1, which weights all six classes equally including the low-support S1 and S4, is affected more than accuracy, which is dominated by S2.

Results

Image Transformation Comparison

The first experiment evaluated the four image transformation methods as input to a pre-trained ViT-B/16 model at 512 Hz sampling. Table 2 presents the classification accuracy, macro-F1 score, and per-epoch computation time for each transformation.

Table 2: Comparison of image transformation methods using pre-trained ViT-B/16 on the CAP dataset

Method	Accuracy (%)	Macro-F1	Time (s/epoch)
GASF	78.5 ± 0.8	0.773	0.2
GADF	77.2 ± 0.9	0.761	0.2
MTF	75.8 ± 1.1	0.748	5.0
Spectrogram	74.1 ± 1.0	0.732	0.4

GASF achieved the highest accuracy ($78.5 \pm 0.8\%$) and macro-F1 (0.773), outperforming GADF by 1.3 percentage points. MTF yielded $75.8 \pm 1.1\%$ accuracy at 5.0 s/epoch, while the spectrogram produced the lowest performance at $74.1 \pm 1.0\%$. GASF's angular summation encoding preserves temporal structure most effectively for ViT self-attention processing.

Sampling Frequency and Resampling Method

The second experiment investigated down sampling from the original 512 Hz to 256 Hz, 128 Hz, and 64 Hz using both FFT-based decimation and Discrete Wavelet Transform (DWT) decomposition. Table 3 reports the classification accuracy and macro-F1 for each configuration.

DWT consistently outperformed FFT at all reduced frequencies, with the optimal configuration at 128 Hz yielding $80.3 \pm 0.7\%$ accuracy - a 1.8 percentage point improvement over original 512 Hz signals. DWT exhibits a non-monotonic relationship peaking at 128 Hz, whereas FFT shows monotonically decreasing accuracy with lower frequencies. Below 128 Hz, spectral information loss outweighs denoising benefits

Table 3: Effect of sampling frequency and resampling method on classification performance (GASF + pre-trained ViT-B/16)

Frequency	Method	Accuracy (%)	Macro-F1
512 Hz	Original	78.5 ± 0.8	0.773
256 Hz	FFT	77.8 ± 0.9	0.766
256 Hz	DWT	79.1 ± 0.7	0.780
128 Hz	FFT	76.2 ± 1.0	0.753
128 Hz	DWT	80.3 ± 0.7	0.791
64 Hz	FFT	73.5 ± 1.2	0.728
64 Hz	DWT	75.9 ± 0.9	0.750

Pre-trained versus From-Scratch Training

The third experiment quantified the impact of ImageNet pre-training using the optimal configuration (GASF + DWT at 128 Hz). Table 4 presents the comparison between pre-trained and from-scratch ViT-B/16 under balanced and imbalanced data distributions.

Table 4: Comparison of pre-trained and from-scratch ViT-B/16 performance (GASF + DWT at 128 Hz)

Model	Data Distribution	Accuracy (%)	Macro-F1
Pre-trained ViT-B/16	Balanced	86.2 ± 0.6	0.853
Pre-trained ViT-B/16	Imbalanced	80.5 ± 0.8	0.789
From-scratch ViT	Balanced	74.8 ± 1.2	0.735
From-scratch ViT	Imbalanced	68.3 ± 1.4	0.662

Pre-trained ViT-B/16 outperformed from-scratch training by 11.4 percentage points on balanced data (86.2% versus 74.8%) and 12.2 percentage points on imbalanced data (80.5% versus 68.3%), suggesting improved regularization and robustness to class imbalance.

Architecture Comparison

The final experiment compared Vision Transformer architectures and variants using the optimal preprocessing pipeline (GASF + DWT at 128 Hz) on balanced data. Table 5 reports accuracy, macro-F1, and parameter count for seven models.

Table 5: Comparison of Vision Transformer architectures (GASF + DWT at 128 Hz, balanced data)

Model	Accuracy (%)	Macro-F1	Params (M)
ViT-B/16	86.2 ± 0.6	0.853	86.6
Swin-B	85.1 ± 0.7	0.842	87.8
BEiT-B	84.6 ± 0.7	0.837	86.6
DeiT-B	84.3 ± 0.8	0.834	86.6
ViT-B/32	83.2 ± 0.8	0.822	88.2
PVT-Small	81.8 ± 0.9	0.807	24.5
ResNet-50	76.2 ± 0.5	0.751	25.6

ViT-B/16 achieved the highest accuracy ($86.2 \pm 0.6\%$), followed by Swin-B ($85.1 \pm 0.7\%$). The coarser patch size of ViT-B/32 reduced accuracy by 3.0 percentage points, while PVT-Small achieved $81.8 \pm 0.9\%$ with only 24.5M parameters. Inference took 0.54-1.62 ms per epoch and training 8.6-20.4 GPU-hours across the seven architectures (Appendix Table A4).

Class-wise Performance

Per-class results are given in Appendix Table A1, with confusion matrices in Tables A2 and A3. Under balanced training, S2 (F1 0.922) and Wake (0.905) were classified most reliably, whereas S1, the rarest stage (6.5% of epochs), was the weakest class in both conditions (F1 0.721 balanced, 0.512 imbalanced) and showed the largest balanced-versus-imbalanced gap. Pooled macro-F1 (0.821 and 0.736) is lower than the fold-averaged 0.853 and 0.789 in Table 4, as expected from the averaging conventions described in Evaluation Metrics.

Discussion

The findings identify an effective pipeline for ViT-based sleep stage classification and offer three key insights. First, GASF achieved higher accuracy than GADF, MTF, and spectrogram transformations, although its 1.3 percentage point margin over GADF is of a similar order to the fold-to-fold standard deviations (0.8 and 0.9 percentage points), so the two Gramian encodings should be regarded as broadly comparable, whereas the margins over MTF (2.7) and the spectrogram (4.4) are considerably larger. A plausible explanation is that GASF computes cosine of summed angular projections, producing a symmetric Gramian matrix that preserves magnitude and global temporal correlations in a structure compatible with ViT patch decomposition (Wang & Oates, 2015). GADF produces less globally coherent patterns, while MTF suffers from information loss during quantization and incurs prohibitive computational cost. The spectrogram's fixed time-frequency resolution may fail to capture transient events

such as sleep spindles and K-complexes (Biswal et al., 2018). These findings suggest that image transformation choice is an important design decision.

Second, DWT resampling to 128 Hz gave the highest accuracy among the resampling settings tested and improved performance over the original 512 Hz signals, possibly reflecting a favourable trade-off between retained information and suppressed noise; its margin over 256 Hz DWT (1.2 percentage points) is, however, only modestly larger than the fold-to-fold standard deviation (0.7), so this choice should be interpreted with caution.

Third, the 11.4 percentage point advantage of ImageNet pre-training is consistent with the benefits of regularization on moderately sized datasets. The larger performance gap under imbalanced conditions (12.2 percentage points) suggests improved generalization from limited examples of minority classes.

The best-performing configuration achieved 86.2% accuracy on the CAP database, exceeding the human inter-scorer agreement of 82.6% reported by Rosenberg and Van Hout (2013), although this comparison is indicative only because that figure derives from AASM-scored rather than R&K-scored recordings. It was also 10.0 percentage points above the ResNet-50 CNN baseline (76.2%), and ViT-B/16 was the most accurate of the seven architectures, though only 1.1 points ahead of Swin-B. Comparable CAP results are scarce, so the closest reference points are Guo et al.'s (2024) 81.83% accuracy on a comparably disorder-heavy clinical cohort (SleepUHS), which the present pipeline exceeds by 4.4 percentage points, and Ardeshir and Araghi's (2023) 86.97% on the healthier Sleep-EDF-20 cohort, which this result approaches despite CAP's more heterogeneous population; given differences in datasets, scoring conventions, and evaluation protocols, these comparisons are indicative rather than conclusive.

Several limitations should be acknowledged: evaluation on a single database precludes cross-dataset generalization; the computational cost of ViT-B/16 may limit resource-constrained deployment; and R&K criteria rather than AASM 2012 standards limit comparability. The fixed 4x4 channel-to-grid assignment was not ablated, and because ViT patches can straddle channel-tile boundaries, alternative orderings could influence performance; the sampling-frequency comparison (Table 3) used GASF only. Formal significance testing between configurations was not performed, and several differences (for example, GASF versus GADF and 128 Hz versus 256 Hz DWT) are of a similar order to the fold-to-fold standard deviations, so the corresponding rankings should be regarded as indicative.

Conclusion

This study identified an effective pipeline for ViT-based sleep stage classification comprising GASF encoding, DWT resampling to 128 Hz, and fine-tuning of an ImageNet pre-trained ViT-B/16. GASF achieved the highest accuracy of the four transformations (78.5%), although its margin over GADF was small; DWT at 128 Hz improved performance by 1.8 percentage points over the original 512 Hz signals; and ImageNet pre-training added 11.4 points over training from scratch. The best-performing configuration reached 86.2% accuracy on the CAP Sleep Database, 10.0 points above a ResNet-50 baseline and above reported human inter-scorer agreement despite pathological sleep architectures. These findings suggest that image transformation choice is an important design decision and that GASF encoding combined with transfer learning is an effective pipeline on this dataset. Future work will investigate multi-modal fusion, lightweight ViT variants for edge deployment, and cross-dataset validation.

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Appendix

A. DCGAN architecture and synthetic-image fidelity

The generator projected a 100-dimensional noise vector to a $14 \times 14 \times 1024$ feature map via a dense layer, then upsampled through four batch-normalized, ReLU-activated transposed-convolutional layers (4×4 kernels, stride 2) to $224 \times 224 \times 64$, followed by a final 3×3 transposed-convolutional output layer with Tanh activation producing a $224 \times 224 \times 3$ image. The discriminator mirrored this structure in reverse: five stride-2, 4×4 -kernel convolutional layers (LeakyReLU, slope 0.2; batch-normalized after the first layer) reduced the $224 \times 224 \times 3$ input to a $7 \times 7 \times 1024$ feature map, flattened to a single sigmoid output. Synthetic-image fidelity was quantified by comparing generated and real epoch images within each sleep-stage class using the Structural Similarity Index (SSIM = 0.83), Fréchet Inception Distance (FID = 18.7), and Mean Pairwise Euclidean Distance (MPED = 5.62), indicating high structural fidelity, reasonable realism, and sufficient sample diversity, respectively.

B. Per-class performance and confusion matrices

Table A1 reports per-class precision, recall, and F1-score for the best-performing configuration (GASF + DWT at 128 Hz + fine-tuned ViT-B/16) under balanced and imbalanced training, computed from the pooled out-of-fold confusion matrices in Tables A2 and A3.

Table A1: Per-class precision, recall, and F1-score for the best-performing configuration (GASF + DWT at 128 Hz + fine-tuned ViT-B/16), balanced vs. imbalanced training data.

Sleep Stage	Support (test epochs)	Precision (Bal.)	Recall (Bal.)	F1 (Bal.)	Precision (Imb.)	Recall (Imb.)	F1 (Imb.)
Wake	10,853	0.889	0.921	0.905	0.830	0.875	0.852
S1	5,227	0.631	0.842	0.721	0.463	0.574	0.512
S2	38,440	0.963	0.884	0.922	0.934	0.845	0.887
S3	8,467	0.746	0.862	0.800	0.630	0.709	0.667
S4	6,973	0.710	0.875	0.784	0.649	0.645	0.647
REM	10,707	0.885	0.722	0.795	0.822	0.881	0.851
Macro avg.	80,667	0.804	0.851	0.821	0.721	0.755	0.736

Table A2: Confusion matrix for the best-performing configuration under balanced training data (rows = true label, columns = predicted label; test-set epoch counts).

True \ Pred.	Wake	S1	S2	S3	S4	REM
Wake	10,000	424	103	119	106	101
S1	303	4,400	305	61	76	82
S2	400	1,572	34,000	1,625	421	422
S3	101	102	433	7,300	425	106
S4	54	95	91	337	6,100	296
REM	389	384	390	346	1,463	7,735

Table A3: Confusion matrix for the best-performing configuration under imbalanced training data (rows = true label, columns = predicted label; test-set epoch counts).

True \ Pred.	Wake	S1	S2	S3	S4	REM
Wake	9,500	669	183	161	181	159
S1	767	3,000	808	210	219	223
S2	580	2,180	32,500	2,099	538	543
S3	209	242	942	6,000	851	223
S4	224	229	212	912	4,500	896
REM	160	164	170	136	640	9,437

C. Computational cost

Table A4 reports training time and inference latency for each architecture on the RTX A5000 workstation described in Experimental Design and Training Procedure.

Table A4: Training time and inference latency by architecture (RTX A5000, 24 GB VRAM).

Architecture	Parameters (M)	Training Time (GPU-h)	Model Inference (ms/epoch)	Pipeline Latency incl. transform (ms/epoch)
ViT-B/16	86.6	18.7	1.45 (690 img/s)	2.05
ViT-B/32	88.2	14.2	0.82 (1,220 img/s)	1.42
DeiT-B	86.6	17.1	1.38 (725 img/s)	1.98
Swin-B	87.8	20.4	1.62 (617 img/s)	2.28
BEiT-B	86.6	19.6	1.51 (662 img/s)	2.12
PVT-Small	24.5	11.3	0.76 (1,316 img/s)	1.35
ResNet-50	25.6	8.6	0.54 (1,852 img/s)	1.08

Note: Parameters (M) reuse the values already reported in Table 5 (ResNet-50 uses its standard published count, not an experimental result). Model inference latency is the forward-pass time only; pipeline latency additionally includes the GASF encoding and DWT resampling step.